
Simulations for Harnessing the VO₂ Phase Transition for Automatic Gain Control in Transimpedance Amplifiers

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1 Yuanhang-based current-drive validation

This run checks that the Yuanhang-based implementation can produce a stable current-driven oscillation with a nonzero voltage valley. It validates the simulation mechanism; it is not a fit to the experimental sample.

All values were taken from Zhang *et al.* [1], except C_{th} . At 600 μA , the nominal value (49.6278 pJ/K) did not sustain oscillations, so it was increased fourfold to slow the thermal response and obtain a stable validation cycle. This control value is not an estimate of our sample's thermal capacitance.

Quantity	Value
Current pulse	600 μA from 2 to 38 μs
Simulation window	0–40 μs
Time step	0.5 ns
Initial state	$V(0) = 0 \text{ V}$, $T(0) = 324.9 \text{ K}$, insulating branch
Electrical capacitance	$C = 145.34619293 \text{ pF}$
Thermal capacitance	$C_{\text{th}} = 198.51107324 \text{ pJ/K}$ ($4\times$ control)
Environmental conductance	$S_e = 0.20558726 \text{ mW/K}$
Ambient temperature	$T_0 = 325 \text{ K}$
Metallic resistance	$R_m = 1286.3165 \Omega$
Noise and thermal neighbor coupling	zero
Analysis interval	10–38 μs

Table 1: Parameters used for the validation run. All unlisted resistance and hysteresis parameters retain their Yuanhang values.

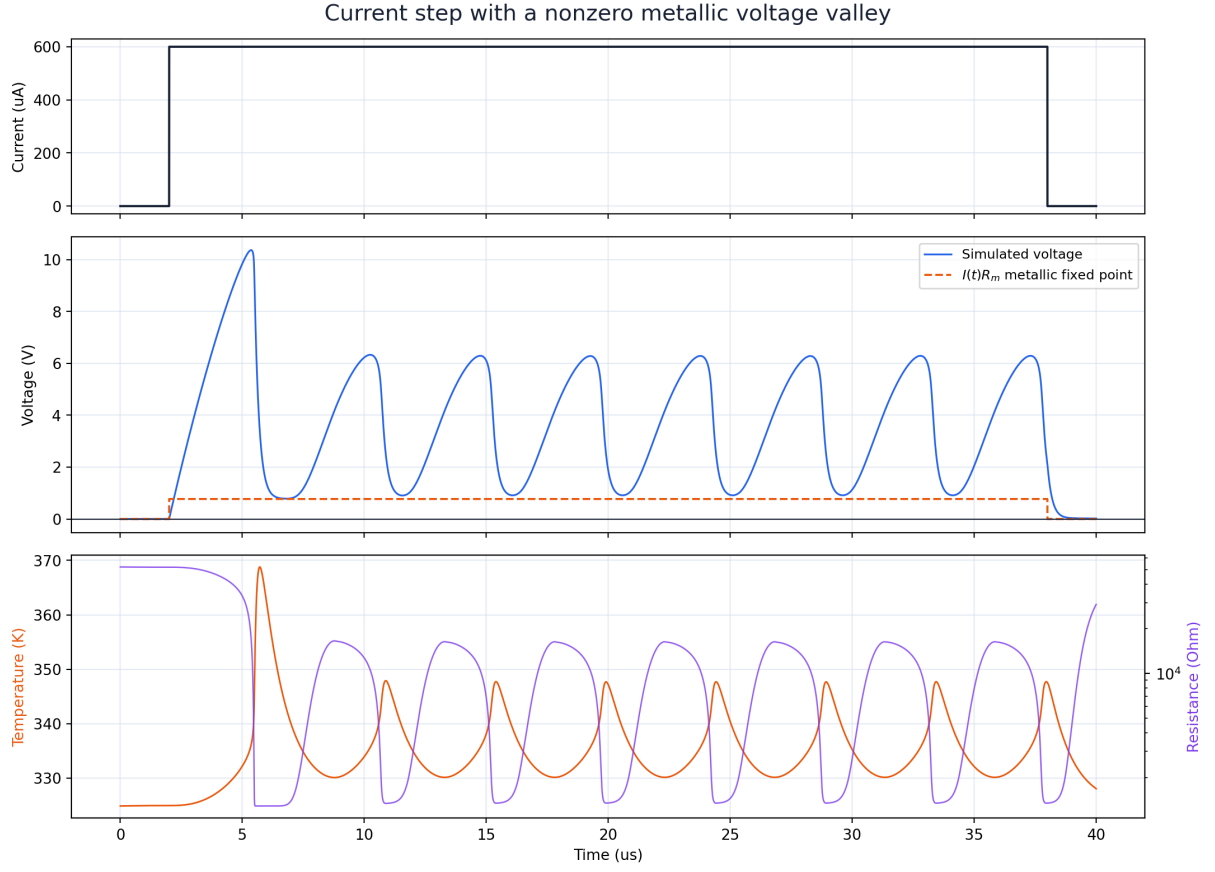


Figure 1: Applied current and simulated voltage, temperature, and resistance. The dashed orange line is the instantaneous metallic fixed point $I(t)R_m$. The analysis excludes the initial charging transient and uses the stable interval from 10 to 38 μs .

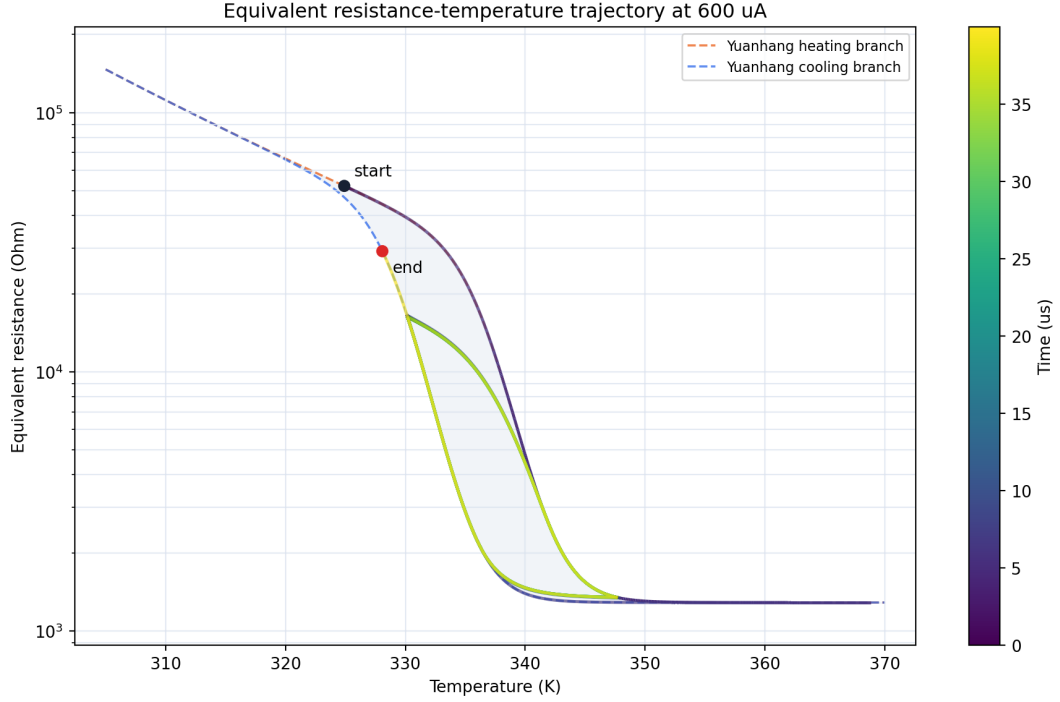


Figure 2: Equivalent resistance versus temperature. The dashed curves and shaded band show Yuanhang’s major heating and cooling hysteresis; color shows progression through the 40 μs simulated trajectory.

Metric	Result
Oscillation detected	yes
Detected cycles	5
Frequency	0.221685 MHz
Period	4.51090 μs
Period coefficient of variation	1.77×10^{-4}
Steady voltage range	0.905654–6.33281 V
Full-pulse temperature range	324.9–368.8 K
Steady temperature range	330.16–347.91 K
Steady resistance range	1.339–16.382 k Ω
Metallic electrical time CR_m	186.961 ns
Thermal time C_{th}/S_e	0.965581 μs

Table 2: Metrics measured from the validation run.

The metallic-state voltage is set approximately by $V = IR_m$:

$$V_{\text{metal}} \approx IR_m = (600 \times 10^{-6} \text{ A})(1286.3165 \Omega) = 0.771790 \text{ V}. \quad (1)$$

The simulated minimum is 0.905654 V; it remains above this fixed point because switching reverses before the capacitor settles. The experimental manuscript reports 40–60 MHz oscillations [2], whereas this validation gives 0.221685 MHz. Sample-specific resistance and time constants are therefore still required for a quantitative match.

2 Sample-specific resistance–temperature fit

The supplied file contains 264 measurements from one cooling sweep and one heating sweep over 289.85–370.16 K at 2 μ A. We fitted the major-loop form

$$R(T, \delta) = R_m + R_0 e^{(E_a/k_B)/T} \left[\frac{1}{2} + \frac{1}{2} \tanh\left(\beta \left(\delta \frac{w}{2} + T_c - T\right)\right) \right], \quad \delta = \begin{cases} -1, & \text{cooling,} \\ +1, & \text{heating.} \end{cases} \quad (2)$$

The fit minimizes the residual in $\log_{10} R$. The minor-loop parameter γ was fixed to Yuanhang’s value because a major loop alone does not determine it.

Here R_0 is the Arrhenius prefactor, not the insulating resistance. On the fully insulating branch,

$$R_{\text{ins}}(T) \approx R_m + R_0 e^{(E_a/k_B)/T}. \quad (3)$$

For example, the fitted sample gives $R_{\text{ins}}(300 \text{ K}) \approx 3.78 \text{ k}\Omega$, well above the metallic floor $R_m = 18.22 \Omega$.

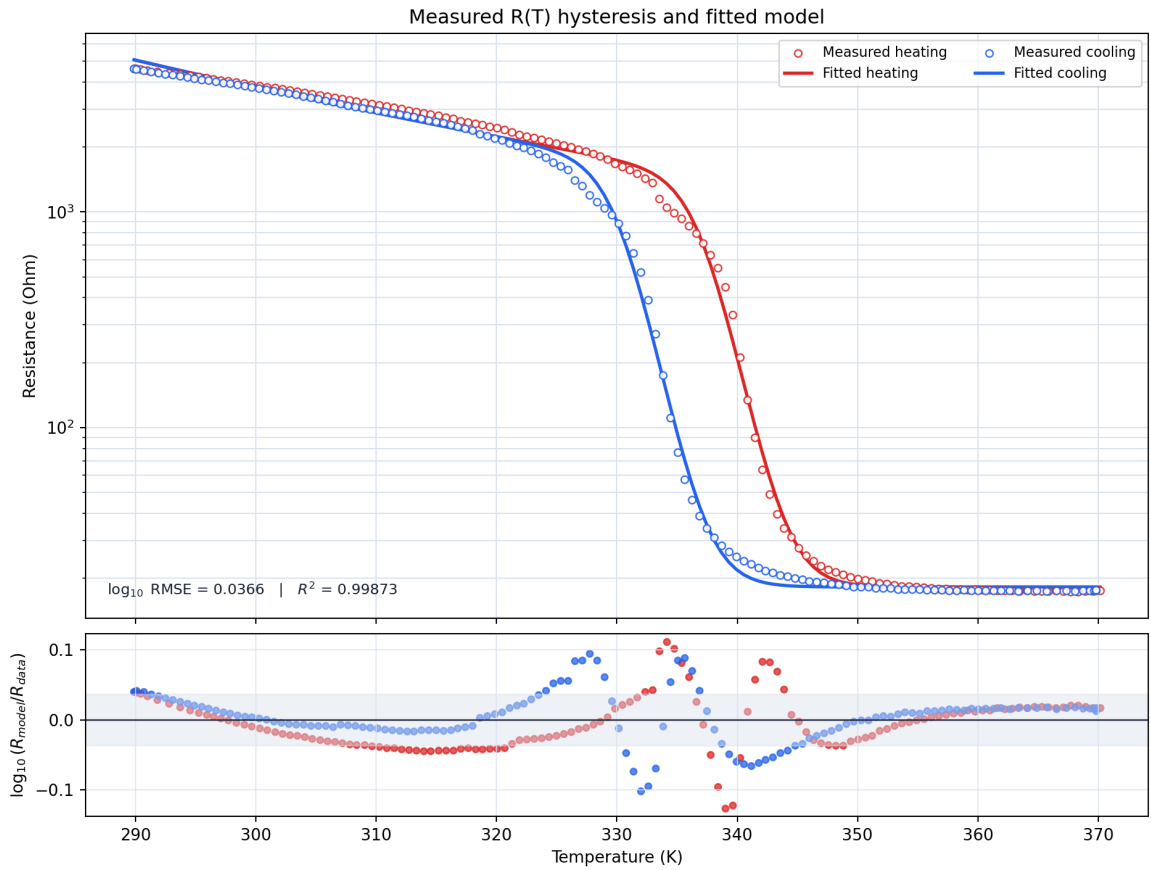


Figure 3: Measured heating and cooling branches, fitted model, and log-resistance residuals.

Parameter	Estimate	95% interval	Yuanhang
R_0 (prefactor)	0.7989 Ω	0.3774–1.8483 Ω	0.005359 Ω
E_a/k_B	2536.97 K	2272.55–2767.52 K	5220.47 K
E_a	0.21862 eV	0.19583–0.23849 eV	0.44987 eV
R_m (metallic floor)	18.2153 Ω	17.4943–18.9010 Ω	1286.32 Ω
T_c	333.494 K	333.137–333.838 K	332.806 K
w	6.8823 K	6.6291–7.2681 K	7.1936 K
β	0.29916 K ⁻¹	0.28092–0.32021 K ⁻¹	0.25280 K ⁻¹
$T_{\text{cool}} = T_c - w/2$	330.052 K	329.642–330.385 K	329.209 K
$T_{\text{heat}} = T_c + w/2$	336.935 K	336.534–337.348 K	336.403 K
γ	0.95627 (fixed)	—	0.95627

Table 3: Sample-specific resistance parameters compared with Yuanhang’s reference implementation, which uses the effective metallic resistance $R_m = 1286.32 \Omega$. R_0 and E_a/k_B are strongly correlated; their joint combination determines the insulating branch.

The fit gives $R^2 = 0.998734$ and a $\log_{10} R$ RMSE of 0.03663, equivalent to a typical multiplicative error of 1.088. The largest deviation is a factor of 1.336 inside the switching region; this residual is retained rather than adding parameters unsupported by the measurement.

3 Experimental timing and ambient temperature

The sample-specific simulations will reproduce the measured pulse timing rather than use an instantaneous current step. The TIA sweep is taken to share the ambient range reported for the related measurements [2].

Input	Value
Ambient temperature	$T_0 \approx 314.4$ K (314.25–314.55 K)
Pulse width	300 ns
Current rise time	26–27 ns (10–90%)
Current fall time	18–23 ns (90–10%)

Table 4: Measured environmental and timing inputs retained for later simulation of the experiment.

4 Environmental thermal conductance

The objective is to estimate the environmental thermal conductance S_e , which sets the heat-loss rate $S_e(T - T_0)$. The estimate requires a hot operating point whose temperature is approximately stationary. We use the following procedure:

1. Order the measured records by their baseline-corrected input current and identify the last record without coherent oscillation. This gives the stable point closest to switching, where the temperature rise is large but a quasi-steady balance remains defensible.
2. Subtract the pre-pulse current and voltage offsets and select a settled window in which V/I is nearly constant. In this window $dV/dt \approx 0$, so the measured input current is approximately the resistive current through the VO₂ device.

3. Calculate the effective resistance $R_* = V_*/I_*$ and Joule power $P_* = I_*V_*$. Invert the fitted heating branch, $R_{\text{heat}}(T_*) = R_*$, to infer the device temperature T_* .
4. Apply the quasi-steady thermal balance $C_{\text{th}} dT/dt = P_* - S_e(T_* - T_0) \approx 0$, which gives $S_e = P_*/(T_* - T_0)$.

The filenames identify the nominal source-voltage setting, not the voltage across the VO₂ device and not the simulation input. We therefore retain that setting only for traceability and use the measured current. Over the common 50–250 ns analysis window, the 250 and 300 mV source settings produced baseline-corrected current steps of 189.6 and 228.2 μA , respectively.

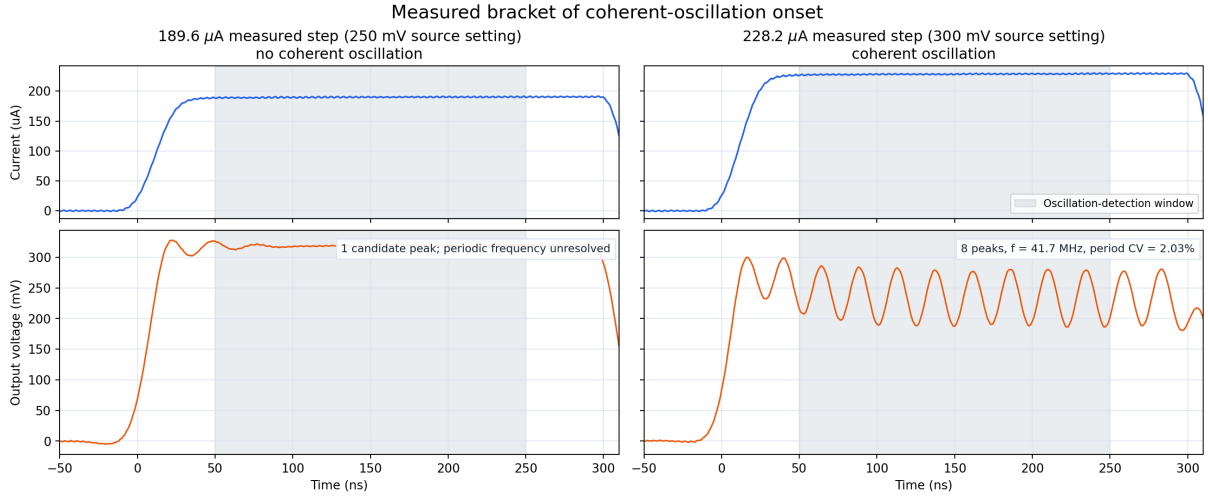


Figure 4: Adjacent baseline-corrected records bracketing oscillation onset on shared axes. The 189.6 μA measured current step (250 mV source setting) contains only one candidate peak, so no periodic frequency is assigned. At 228.2 μA (300 mV source setting), eight peaks give 41.7 MHz with a 2.03% period coefficient of variation.

Figure 4 implements the first step of the procedure. The 228.2 μA record is the first trace with coherent oscillation, while the immediately preceding 189.6 μA record does not oscillate. We therefore selected the latter as the closest measured point below switching onset. We did not use the first transition during the current edge because the device is still heating there and $dT/dt = 0$ would not be justified.

For the selected trace, the current and voltage offsets are their medians from -200 to -50 ns. We then use the settled 100–250 ns interval. Its effective resistance changes by only 0.145%, supporting the quasi-steady approximation. Figure 5 shows the waveform, the stability test, and the temperature inversion.

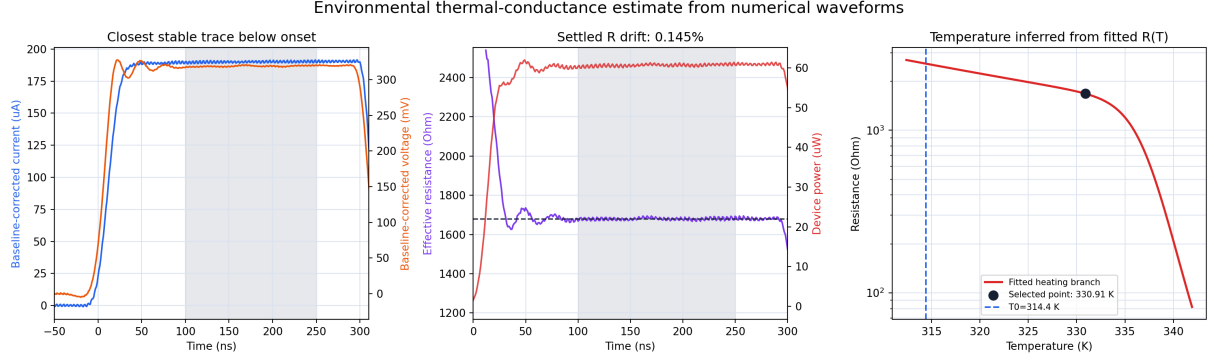


Figure 5: Environmental-conductance evidence. Gray shading marks the settled interval. The measured $R_* = V_*/I_*$ is mapped to T_* through the fitted heating branch.

The baseline-corrected medians give

$$I_* = 190.162 \mu\text{A}, \quad V_* = 319.213 \text{ mV}, \quad R_* = \frac{V_*}{I_*} = 1679.759 \Omega, \quad (4)$$

and

$$P_* = I_* V_* = 60.658 \mu\text{W}. \quad (5)$$

Solving the fitted heating branch $R_{\text{heat}}(T_*) = R_*$ gives $T_* = 330.905 \text{ K}$. At a quasi-steady point, the thermal equation reduces to

$$C_{\text{th}} \frac{dT}{dt} = P_* - S_e(T_* - T_0) \approx 0, \quad S_e = \frac{P_*}{T_* - T_0}. \quad (6)$$

Therefore,

$$S_e = \frac{60.658 \mu\text{W}}{330.905 \text{ K} - 314.4 \text{ K}} = 3.675 \mu\text{W/K} = 0.003675 \text{ mW/K}. \quad (7)$$

Result	Value
Selected trace	190.162 μA settled current (250 mV setting)
First oscillating trace	228.2 μA current step (300 mV setting)
Settled resistance drift	0.145%
Inferred temperature	330.905 K
Temperature rise	16.505 K
S_e estimate	0.003675 mW/K
Conditional 95% interval	0.003434–0.004085 mW/K
Yuanhang reference	0.205587 mW/K

Table 5: Environmental thermal-conductance calculation and comparison.

The interval propagates paired waveform block resampling, the 1000-sample $R(T)$ fit bootstrap, and the 314.25–314.55 K ambient range. Omitting baseline correction gives 0.003560 mW/K, inside the interval. The result remains conditional on the $R(T)$ curve and TIA traces belonging to the same device and on the quasi-static heating branch applying under electrical drive. Yuanhang’s conductance is about 56 times larger.

5 Electrical capacitance

The objective is to estimate the effective electrical capacitance C , which stores charge at the device node and sets the electrical response time $\tau_{\text{el}} = R(T)C$. In the ideal-current model, the measured input current splits between the VO₂ resistance and this capacitance:

$$I_{\text{in}} = \frac{V}{R(T)} + C \frac{dV}{dt}. \quad (8)$$

We therefore seek the part of the measured current that cannot be explained by the instantaneous resistive current $V/R(T)$. The analysis uses low-current, non-switching rising edges so that the device begins on the known cold heating branch and a phase transition does not complicate the electrical transient. It assumes that the current and voltage channels are time-aligned, their pre-pulse offsets are removable, C is approximately constant over the fit window, and the reduced circuit contains the relevant current paths. The procedure is:

1. Subtract each trace’s pre-pulse current and voltage medians and select the initial rising edge of the 100–250 mV source-setting records, all of which remain below switching.
2. Use $C_{\text{shortcut}} = \Delta I / (dV/dt)$ as a preliminary scale estimate. This expression is valid only if the resistive current is negligible; it must not be accepted before checking the full current balance.
3. Evaluate the physically complete residual $I_C = I_{\text{in}} - V/R(T)$ using the fitted cold resistance, and fit $I_C = C dV/dt$ over the early edge with the physical constraint $C \geq 0$.
4. Check the result independently against the measured current–voltage timing. If no positive lag is resolved, convert the timing resolution Δt into an upper bound $C \lesssim \Delta t / R(T_0)$. We adopt that bound as a conservative modeling value while retaining $C = 0$ as the best-fit lower endpoint.

We first reproduce the preliminary shortcut. For each trace, ΔI is the baseline-to-plateau current step and dV/dt is a linear fit over 0–30 ns, giving

$$C_{\text{shortcut}} = \frac{\Delta I}{dV/dt}, \quad \frac{\mu\text{A}}{\text{mV/ns}} = \text{pF}. \quad (9)$$

Drive setting	ΔI (μA)	dV/dt (mV/ns)	C_{shortcut} (pF)
100 mV	76.171	3.562	21.382
150 mV	114.172	6.127	18.634
200 mV	152.162	8.235	18.477
250 mV	189.601	9.008	21.047
Median			19.841

Table 6: Reconstruction of the original 19.8 pF shortcut estimate.

The median shortcut estimate is 19.8 pF, but it is biased because the measured current itself rises over 26–27 ns and most of it flows resistively. Applying the full current balance gives

$$C \frac{dV}{dt} = I_{\text{in}} - \frac{V}{R(T)}, \quad C = \frac{I_{\text{in}} - V/R(T)}{dV/dt}. \quad (10)$$

At $T_0 = 314.4$ K, the fitted heating branch gives $R(T_0) = 2.570$ k Ω . Substituting the measured current and voltage leaves no resolved capacitive current: the unconstrained early-edge fit is approximately -0.25 to -0.10 pF, depending on the fit window, and the physical nonnegative optimum is therefore $C = 0$. The current and voltage 50% crossings differ by only -0.74 to $+0.52$ ns. Taking 1 ns as the timing resolution gives the non-statistical bound

$$C \lesssim \frac{1 \text{ ns}}{2.570 \text{ k}\Omega} \approx 0.39 \text{ pF}. \quad (11)$$

In contrast, 19.8 pF would give $R(T_0)C = 50.9$ ns and a large voltage lag that is absent from the data. A larger capacitance would worsen the disagreement. This does not establish that the physical capacitance is exactly zero; it means that a positive value is not identifiable from the present traces. To avoid assuming an instantaneous electrical response, the sample-specific simulations and downstream thermal fit use the conservative upper-bound value

$$C_{\text{adopt}} = 0.39 \text{ pF}, \quad I_R = I_{\text{in}} - C_{\text{adopt}} \frac{dV}{dt}. \quad (12)$$

This is not a measured central estimate: $C = 0$ remains the constrained best fit, and 0.39 pF intentionally maximizes the electrical memory still compatible with the approximately 1 ns timing resolution.

This conclusion is independent of the present choice of γ . The capacitance fit uses the initial cold heating edge, before a temperature reversal; γ modifies only minor hysteresis loops after a reversal and therefore does not enter this calculation. Changing γ can alter later oscillation shapes and frequencies, but it cannot supply the missing early-edge voltage lag. A numerical nonzero estimate requires a faster, low-amplitude nonswitching edge with calibrated current-voltage channel timing; with R approximately constant, $C = \tau_{\text{el}}/R$.

6 Thermal time constant and thermal capacitance

The objective is to estimate the effective thermal capacitance C_{th} , which relates net heating power to the device's temperature rate of change. Together with the environmental conductance found in Section 4, it sets the thermal response time

$$\tau_{\text{th}} = \frac{C_{\text{th}}}{S_e}. \quad (13)$$

A larger C_{th} means that more energy is required to change the device temperature and therefore produces a slower thermal response. Section 5 found no resolved positive electrical capacitance but established the upper bound $C_{\text{adopt}} = 0.39$ pF. We use this conservative value and first separate the resistive current from the capacitive current. The fitted heating branch then acts as a thermometer:

$$I_R(t) = I_*(t) - C_{\text{adopt}} \frac{dV_*}{dt}, \quad R_*(t) = \frac{V_*(t)}{I_R(t)}, \quad T_*(t) = R_{\text{heat}}^{-1}(R_*(t)). \quad (14)$$

The estimate assumes a single lumped device temperature, constant C_{th} and S_e over the fitted interval, the same sample for the electrical and $R(T)$ measurements, and validity of the fitted heating branch before any thermal reversal. It is conditional on the adopted 0.39 pF upper bound and requires baseline-corrected, time-aligned current and voltage channels. The procedure is:

1. Select moderate non-switching traces with sufficient signal, subtract their pre-pulse offsets, and smooth the oscilloscope noise before evaluating dV_*/dt and forming V_*/I_R . We use measured current steps of 76.2, 114.2, and 152.2 μA (the 100, 150, and 200 mV source settings); the 189.6 μA record (250 mV setting) is reserved as a near-transition sensitivity check.
2. Compute $I_R = I_* - C_{\text{adopt}}dV_*/dt$, reconstruct the measured temperature trajectory from $R_*(t) = V_*(t)/I_R(t)$ and the inverse fitted heating branch, and calculate the applied Joule power $P_*(t) = I_R(t)V_*(t)$.
3. For trial values of C_{th} , integrate the thermal balance using the previously estimated S_e and T_0 . Fit one shared C_{th} by minimizing the temperature mismatch across all three primary traces over 15–35 ns, then obtain $\tau_{\text{th}} = C_{\text{th}}/S_e$.
4. Test robustness by fitting each trace separately, checking the excluded 189.6 μA record, varying the fit window, and propagating the $R(T)$, S_e , ambient temperature, and waveform uncertainties.

The thermal trajectory for each trial value is obtained from

$$C_{\text{th}} \frac{dT}{dt} = P_*(t) - S_e(T - T_0), \quad C_{\text{th}} = S_e \tau_{\text{th}}. \quad (15)$$

A nine-sample, second-order Savitzky–Golay filter is used in the reconstruction. Because the fit ends before a thermal reversal, γ does not enter this estimate. Figure 6 compares the temperatures inferred directly from the electrical measurements with the trajectories predicted by the shared thermal model.

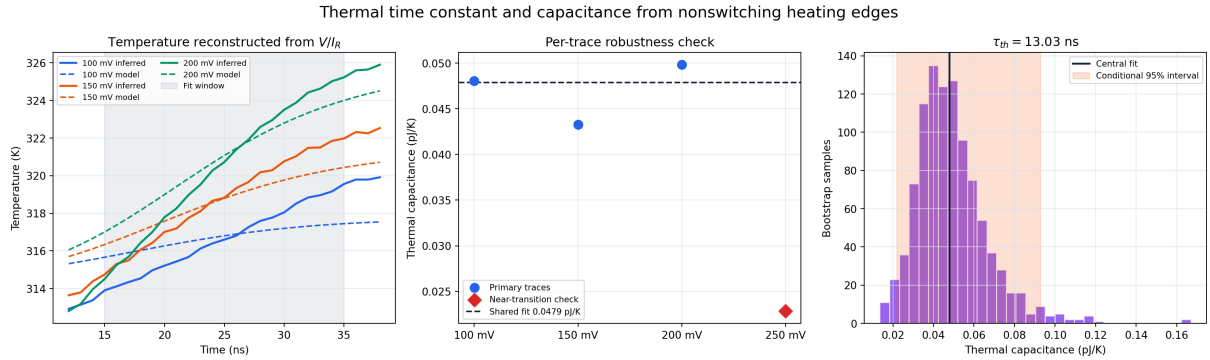


Figure 6: Thermal-capacitance evidence conditioned on $C_{\text{adopt}} = 0.39$ pF. Solid curves are temperatures inferred from V_*/I_R ; dashed curves are the shared thermal model. The 189.6 μA measured current step (250 mV source setting) is retained only as a near-transition sensitivity check.

Quantity	Result
Ambient temperature	314.4 K
Environmental conductance	0.00367513 mW/K
Adopted electrical capacitance	0.39 pF (upper bound)
Selected traces	76.2, 114.2, and 152.2 μ A steps
Fit interval	15–35 ns
Thermal time constant τ_{th}	13.026 ns
Conditional 95% interval for τ_{th}	5.865–25.174 ns
Thermal capacitance C_{th}	0.047873 pJ/K
Conditional 95% interval for C_{th}	0.021918–0.092624 pJ/K
Temperature-fit RMSE	1.145 K
Individual selected-trace range	0.043261–0.049830 pJ/K
189.6 μ A sensitivity result	0.022858 pJ/K; RMSE 2.380 K
Measured oscillation period	16–24 ns
Yuanhang reference	49.6278 pJ/K

Table 7: Thermal time-constant and capacitance results.

The conditional interval propagates trace resampling, the $R(T)$ bootstrap, the conductance bootstrap, the ambient-temperature range, and ± 2 ns changes to the fit window. The archived conductance samples do not retain their paired $R(T)$ parameter draws, so the two bootstraps are resampled independently; this is conservative but does not preserve their covariance. The result remains conditional on the adopted electrical-capacitance upper bound, the static heating branch describing the driven device, and the electrical and $R(T)$ measurements belonging to the same sample. Relative to the $C = 0$ analysis, the upper-bound correction raises the central C_{th} estimate by only 1.5% (0.047163 to 0.047873 pJ/K), although the RMSE increases from 0.806 to 1.145 K. The 189.6 μ A trace gives a smaller value and a substantially larger error because it approaches a reversal; including it would bias the single-heating-branch fit downward. The fitted 13.0 ns thermal time is of the same order as the measured 16–24 ns oscillation period. Yuanhang’s thermal capacitance is approximately 1037 times larger.

The experimental manuscript reports a 150 nm film thickness and an approximately 200 nm electrode gap [2], but not the electrically active filament width or volume. A numerical geometry estimate $C_{\text{th}} = \rho c_p V$ would therefore introduce an unconstrained volume and is not reported.

7 Comparison across experimental currents

This is a blind prediction test: the fitted $R(T)$ curve, $T_0 = 314.4$ K, $S_e = 0.00367513$ mW/K, $C = 0.39$ pF, and $C_{th} = 0.047873$ pJ/K are fixed before examining the result. Each baseline-corrected measured current waveform is replayed through the model; no parameter is adjusted by current. Predicted voltages are sampled at the experimental 1 ns times, and both data sets use the same 50–250 ns window and oscillation detector.

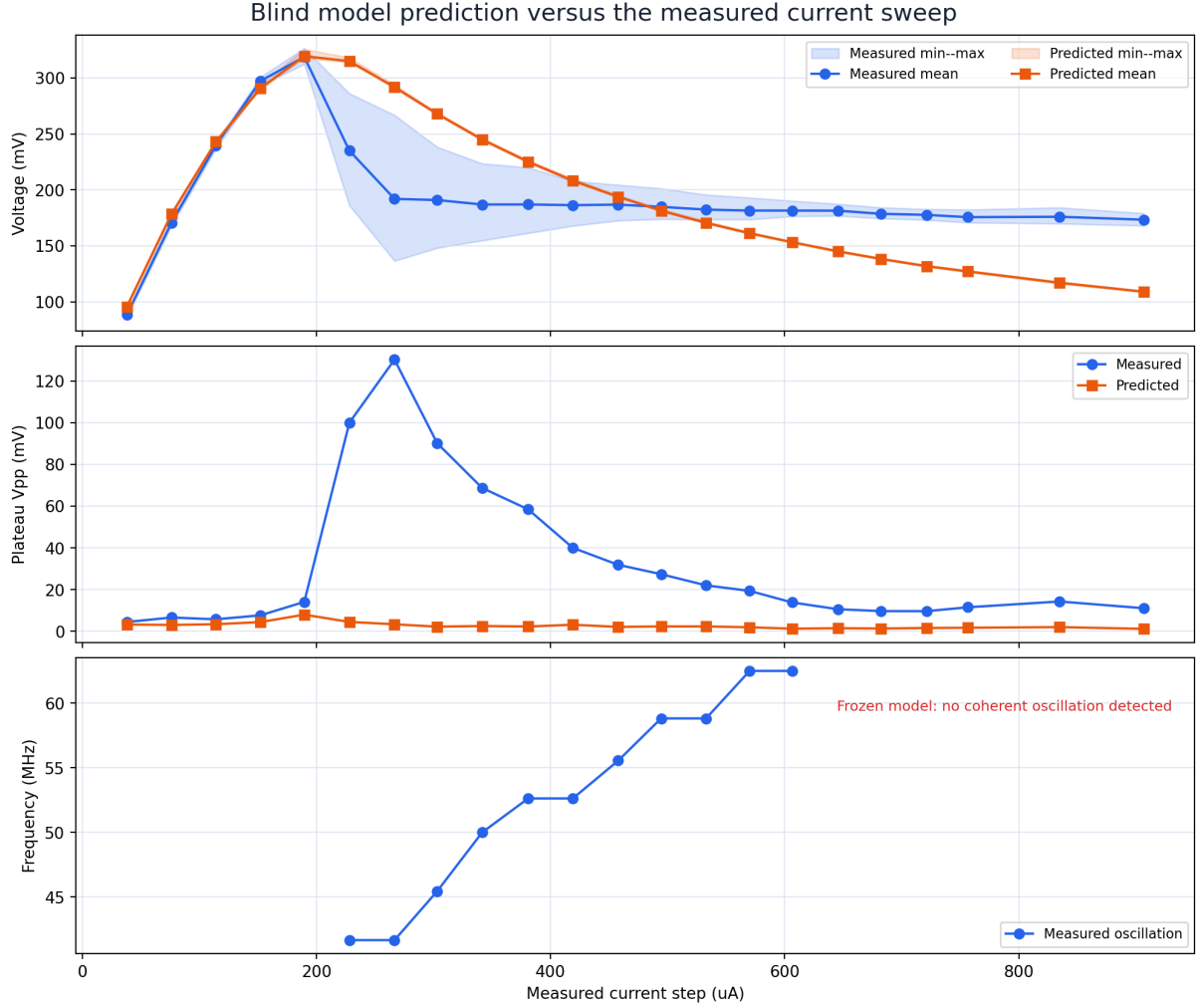


Figure 7: Measured behavior and blind model predictions over all 22 measured current steps. The shaded bands show the voltage extrema in the common analysis window.

The experiment oscillates over 11 records, from 228.2 to 606.3 μA , with frequencies of 41.7–62.5 MHz. The frozen model predicts no coherent oscillation at any measured current. It nevertheless reproduces the last stable pre-onset record: at 189.6 μA , the measured and predicted mean voltages are 318.86 and 319.53 mV. Thus the cold-side static calibration works, whereas the dynamic switching window does not.

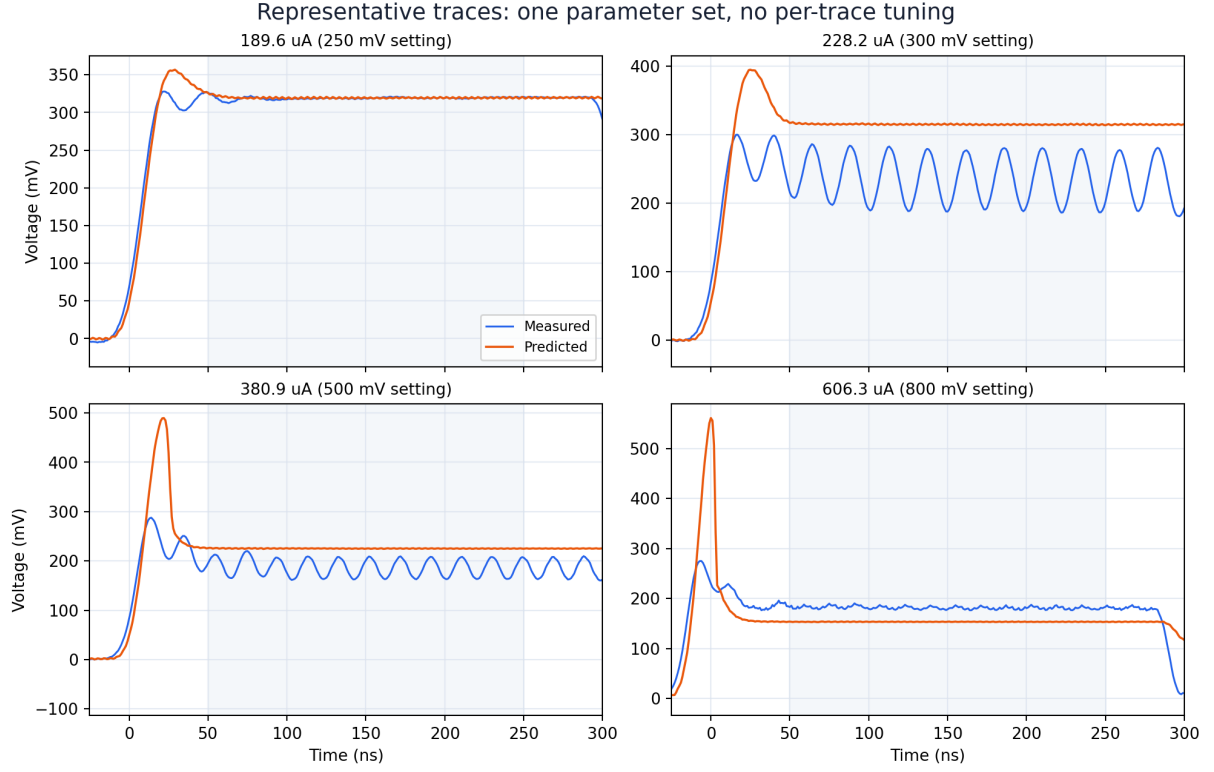


Figure 8: Representative measured and predicted voltages. The shaded interval is the common detection window. The 189.6 μA control is reproduced closely, but the measured oscillations above onset are replaced by steady model solutions.

Quantity	Experiment	Frozen model
Oscillating records	11 of 22	0 of 22
Oscillation-current interval	228.2–606.3 μA	none
Frequency interval	41.7–62.5 MHz	—
Energy per cycle in oscillating records	1.229–1.761 pJ	—
189.6 μA mean voltage	318.86 mV	319.53 mV
Median plateau-voltage RMSE	32.38 mV	
Maximum change after halving Δt	0.228 mV	

Table 8: Direct prediction results. Energy per cycle is the mean plateau power divided by the detected frequency.

Halving the integration step from 0.05 to 0.025 ns does not change any tested classification and changes the representative mean voltages by at most 0.228 mV. The missing oscillations are therefore not a time-step artifact.

8 Capacitance dependence and the thermal-only limit

In the zero-capacitance limit,

$$C = 0 \implies V(t) = I_{\text{in}}(t)R(T), \quad \tau_{\text{th}} = \frac{C_{\text{th}}}{S_e}. \quad (16)$$

A thermal relaxation cycle requires the same current to heat through the heating transition and then cool through the cooling transition. Using the fitted branch resistances at the two transition centers gives

$$I > I_{\text{heat}} = \sqrt{\frac{S_e(T_{\text{heat}} - T_0)}{R_{\text{heat}}}} = 329.65 \mu\text{A}, \quad I < I_{\text{cool}} = \sqrt{\frac{S_e(T_{\text{cool}} - T_0)}{R_{\text{cool}}}} = 254.44 \mu\text{A}. \quad (17)$$

Because $I_{\text{heat}} > I_{\text{cool}}$, these conditions do not overlap: the fitted system has no thermal-only oscillation window. At the adopted $C = 0.39$ pF, the metallic electrical time is $CR_m = 7.10$ ps, compared with $\tau_{\text{th}} = 13.026$ ns. The ratio is 5.45×10^{-4} , so this capacitance supplies negligible electrical memory and does not alter the steady thermal thresholds.

The numerical study varies C from 0 to 50 pF and C_{th} across its conditional 0.021918–0.092624 pJ/K interval while holding every other parameter fixed. The experimentally admissible range $0 \leq C \leq 0.39$ pF produces no oscillation anywhere in that interval. Oscillations first appear at 3 pF only at the lowest tested C_{th} ; at the adopted C_{th} , they first appear at 7 pF. Both values exceed the electrical timing bound and would create a voltage lag absent from the measured edge.

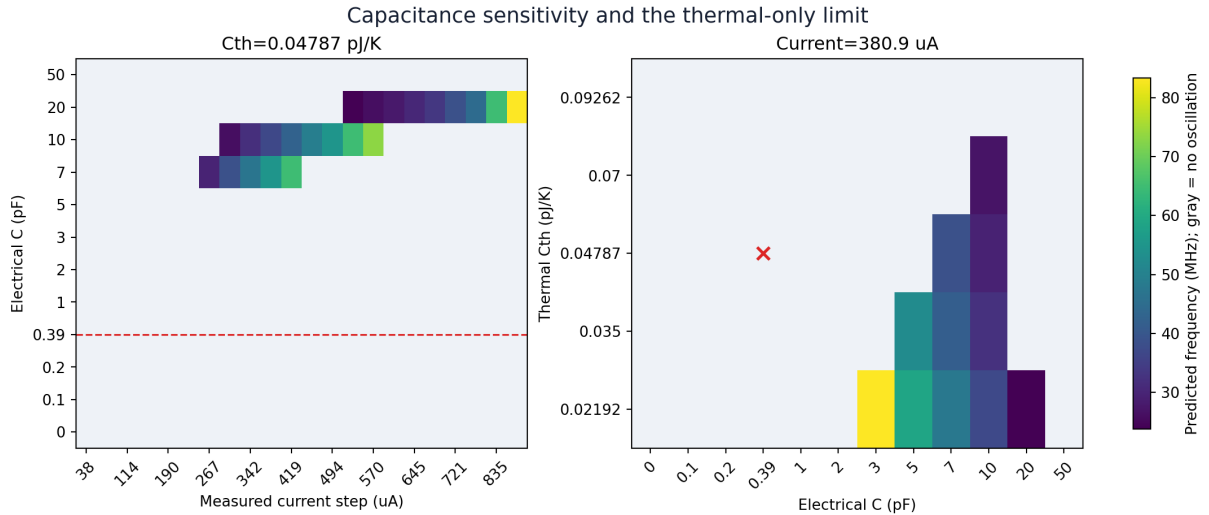


Figure 9: Predicted frequency under capacitance variation. Gray denotes no coherent oscillation. The dashed line and red cross identify the adopted electrical and thermal capacitances; larger electrical values are stress tests outside the timing bound.

The present parameter set therefore matches the stable pre-onset voltage but cannot reproduce the measured oscillatory regime without an electrical capacitance that contradicts the edge timing. The next useful tests are an independently measured dynamic switching loop and inclusion of the actual TIA/load impedance. Changing γ alone cannot explain the onset at $228.2 \mu\text{A}$, because the predicted temperature there does not reach the first heating transition; γ acts only after a hysteresis reversal.

9 Global parameter inference from all measured waveforms

We next asked whether one shared set of model parameters could be inferred directly from the complete current–voltage data set. This is a stricter test than fitting one trace: the same vector

$$\theta = (C, C_{\text{th}}, S_e, T_0, T_c, w, \beta, \gamma) \quad (18)$$

must describe all 22 measured current waveforms. The source settings at 200, 400, 600, 800, and 1000 mV were withheld from optimization, leaving 17 training traces and five blind test traces. The source-setting labels only identify records; the measured current waveform is the model input in every case.

For each candidate, the simulator uses the complete measured current transient and compares the predicted and measured voltage over the common 50–250 ns plateau. The weighted objective is

$$J = 0.20J_{\text{wave}} + 0.20J_{\text{mean}} + 0.15J_{\text{amp}} + 0.15J_{\text{spec}} + 0.15J_f + 0.10J_{\text{class}} + 0.05J_{\text{edge}} + 0.05J_{\text{prior}}. \quad (19)$$

Here J_{wave} is a normalized waveform error, with a ± 12 ns phase shift allowed only for oscillatory records; the other terms compare mean voltage, peak-to-peak amplitude, normalized spectrum, frequency, oscillation classification, and the 0–50 ns rising edge. The constrained fit includes a weak prior and keeps each parameter inside its independently estimated interval. A second, deliberately relaxed fit removes that prior and expands the bounds to test whether the model can fit the data only by abandoning those measurements. Differential evolution explores the global space, followed by bounded local refinement. Optimization uses $\Delta t = 0.2$ ns; every reported candidate is rerun at 0.05 ns.

Parameter	Starting value	Physical interval	Constrained fit	Relaxed fit
C (pF)	0.3900	0–0.390	0.2408	5.4187*
C_{th} (pJ/K)	0.04787	0.02192–0.09262	0.06146	0.10643*
S_e (mW/K)	0.003675	0.003434–0.004085	0.003621	0.004894*
T_0 (K)	314.400	314.25–314.55	314.416	317.844*
T_c (K)	333.494	333.137–333.838	333.476	332.927*
w (K)	6.8823	6.6291–7.2681	6.8726	2.5568*
β (K ^{−1})	0.29916	0.28092–0.32021	0.29542	0.11613*
γ	0.95627	0.2–2.0	0.97956	1.03546

Table 9: Shared parameters inferred from all training traces. An asterisk marks a relaxed value outside the independently supported physical interval.

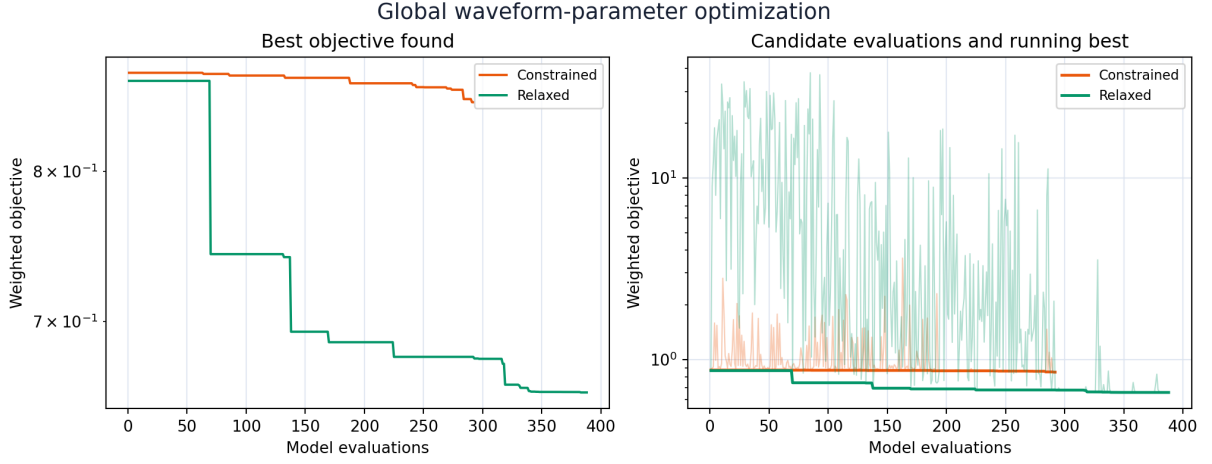


Figure 10: Optimization history for the physically constrained and relaxed searches. The large relaxed improvement occurs only after the search leaves the measured parameter region.

Model	Training J	Held-out J	All-trace J	Oscillations
Starting estimates	0.8680	0.9267	0.8813	0 of 22
Constrained fit	0.8601	0.9255	0.8750	0 of 22
Relaxed diagnostic	0.6515	0.7888	0.6827	0 of 22

Table 10: Fit and blind-test performance. Lower objective values are better. The experiment contains 11 oscillatory records.

The constrained fit lowers the all-trace objective by only 0.72% and the held-out objective by 0.13%. It remains essentially the independently estimated model and still predicts no oscillations. The relaxed search lowers the all-trace objective by 22.5% and the held-out objective by 14.9%, so its improvement is not confined to the training records. However, seven of its eight parameters lie outside their physical intervals, including $C = 5.42$ pF, and it also predicts no coherent oscillation. Its reduced amplitude error comes from large turn-on transients that settle to steady voltages.

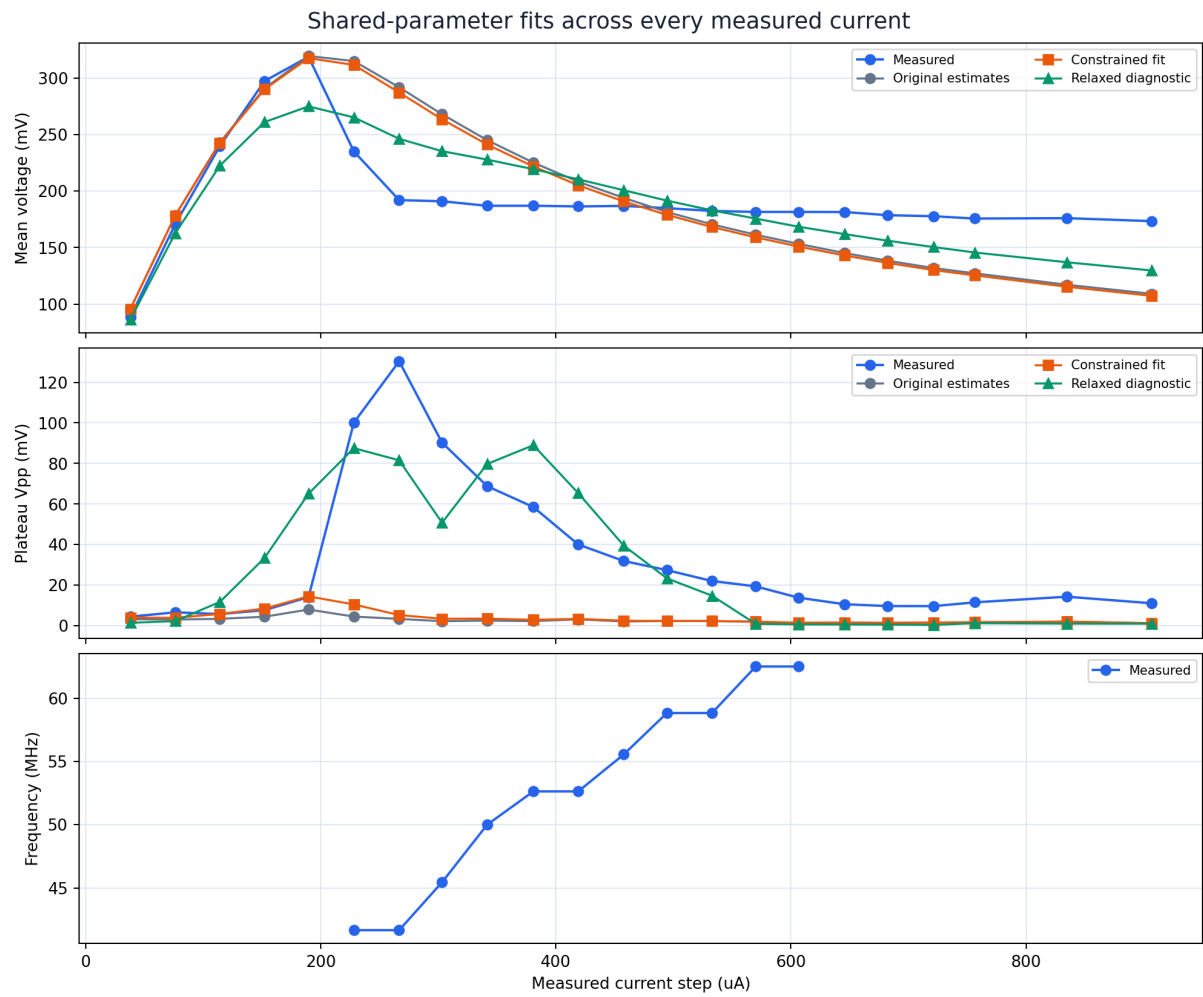


Figure 11: Measured and fitted operating summaries. The relaxed search partially reproduces mean voltage and apparent amplitude, but neither fit produces the measured frequency branch.

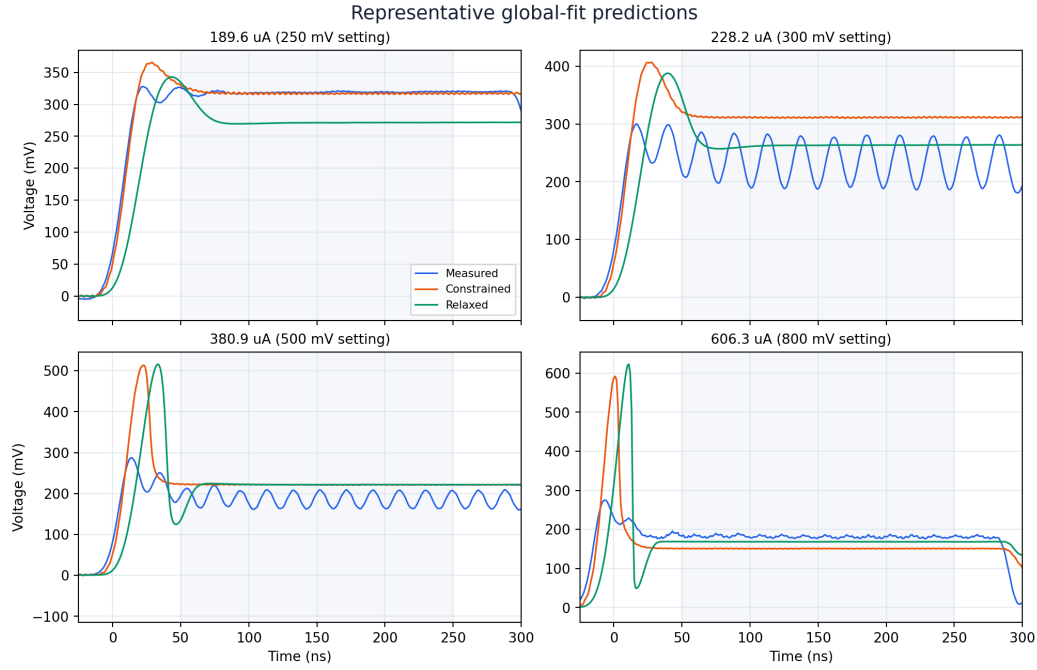


Figure 12: Representative waveform predictions. The relaxed fit can create a large initial excursion, but both fitted models settle instead of sustaining the measured oscillations.

Repeating the final evaluation at 0.2, 0.1, and 0.05 ns gives all-trace objectives of 0.6868, 0.6838, and 0.6827 for the relaxed fit, without changing any oscillation classification. The conclusion is therefore numerically converged. This exercise does not provide replacement physical parameter estimates: it shows that parameter adjustment within the present ideal-current, quasi-static-hysteresis model is insufficient. The likely missing information is dynamic switching behavior or the actual TIA/load impedance. Notably, γ remains near Yuanhang’s value in both searches, so changing the minor-loop curvature alone is not the solution.

10 Oscillation-priority parameter inference

The balanced objective in Section 9 rewards average waveform agreement, so the optimizer can prefer a turn-on transient over sustained oscillation. We therefore repeated the shared-parameter fit with oscillation tracking as the first priority. A sinusoid at each measured oscillation frequency is fitted separately in four consecutive plateau segments. A candidate receives persistence credit only when that periodic component remains present throughout the plateau.

The final objective is

$$J = 250J_{\text{class}} + 250J_{\text{false+}} + 0.50J_f + 0.50J_{\text{sustain}} + 0.05J_{\text{spec}} + 0.02J_{\text{wave}} + 0.02J_{\text{mean}} + 0.01J_{\text{amp}} + 0.01J_{\text{edge}} + 0.01J_{\text{prior}}. \quad (20)$$

Thus one classification error dominates all ordinary waveform terms. A false oscillation outside the measured window contributes to both J_{class} and $J_{\text{false+}}$, so it costs twice as much as missing one boundary record. The search was performed directly at $\Delta t = 0.05$ ns. The development sequence was:

1. A 0.2 ns scouting fit produced nine apparent oscillatory records, but only three survived at 0.05 ns; this candidate was rejected.
2. A direct 0.05 ns fit produced five consecutive oscillatory records and classified 16 of 22 records correctly.
3. A full-resolution C – C_{th} neighborhood identified a training-set seed at $C = 8$ pF and $C_{\text{th}} = 0.020$ pJ/K.
4. Differential evolution and bounded local refinement optimized the remaining secondary errors while preserving the best classification basin.

The final optimization still excludes the five source settings used previously for testing. Because those records had already been inspected during earlier model development, their results are useful withheld-current checks but are not claimed as a pristine blind test.

Parameter	Independently estimated value	Physical interval	Oscillation fit
C (pF)	0.3900	0–0.390	13.8147*
C_{th} (pJ/K)	0.04787	0.02192–0.09262	0.01626*
S_e (mW/K)	0.003675	0.003434–0.004085	0.002736*
T_0 (K)	314.400	314.25–314.55	317.538*
T_c (K)	333.494	333.137–333.838	332.844*
w (K)	6.8823	6.6291–7.2681	12.7734*
β (K ^{−1})	0.29916	0.28092–0.32021	0.39218*
γ	0.95627	0.2–2.0	2.65135*

Table 11: Classification-first shared fit. Every inferred value lies outside its independently supported physical interval.

Quantity	Experiment	Oscillation-priority fit
Correct classifications	22 reference labels	21 of 22
Oscillating records	11	10
Oscillation-current interval	228.2–606.3 μA	228.2–570.1 μA
Frequency interval	41.7–62.5 MHz	25.6–50.0 MHz
Training classifications	17 of 17	17 of 17
Withheld-current classifications	5 reference labels	4 of 5
False-positive oscillations	0	0

Table 12: Oscillation tracking at $\Delta t = 0.05$ ns. The only classification error is the withheld 606.3 μA boundary record.

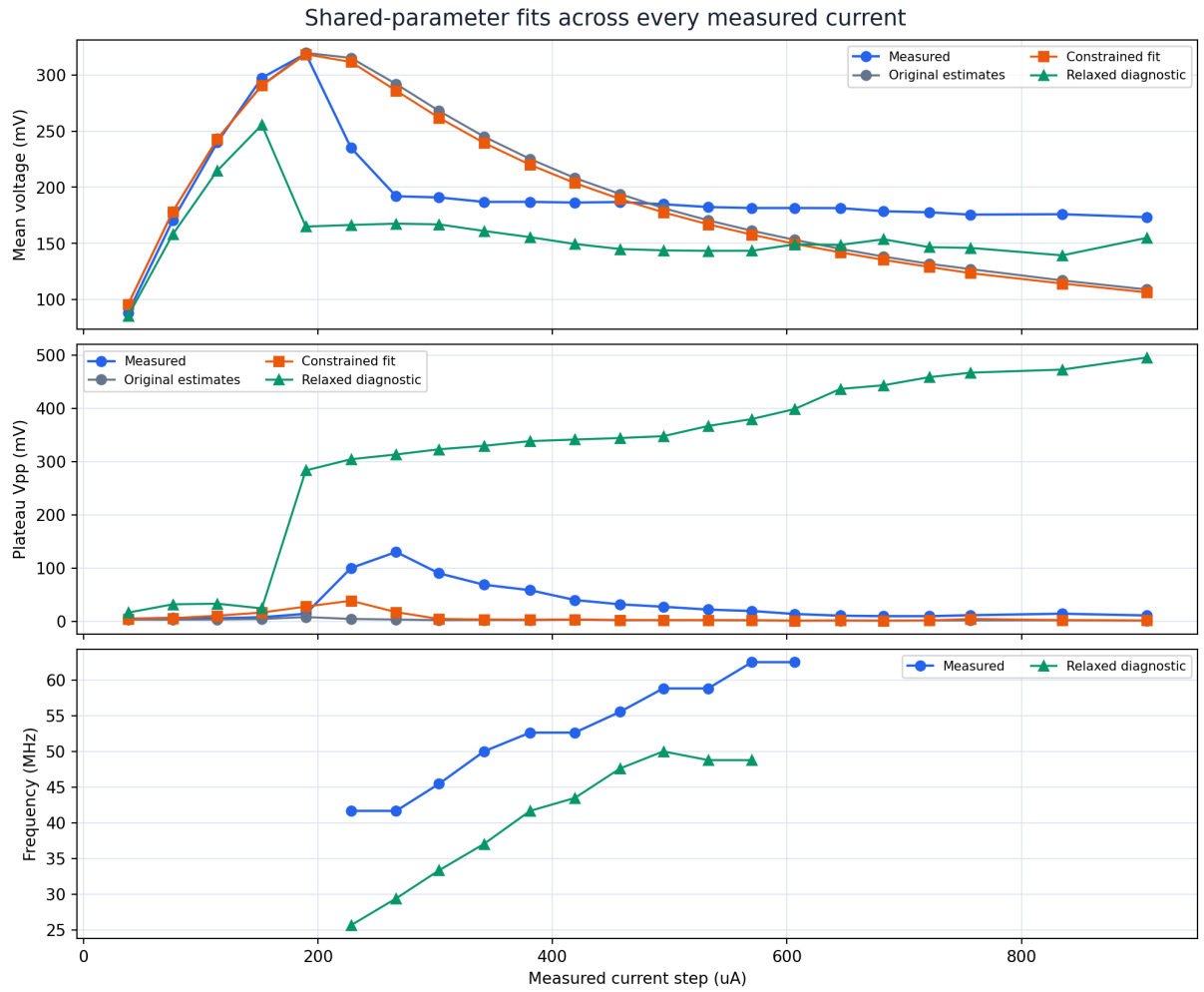


Figure 13: Measured and classification-first operating behavior. The model now reproduces most of the oscillation window and an increasing frequency trend, but its cycle amplitude is much too large and its frequency is generally too low.

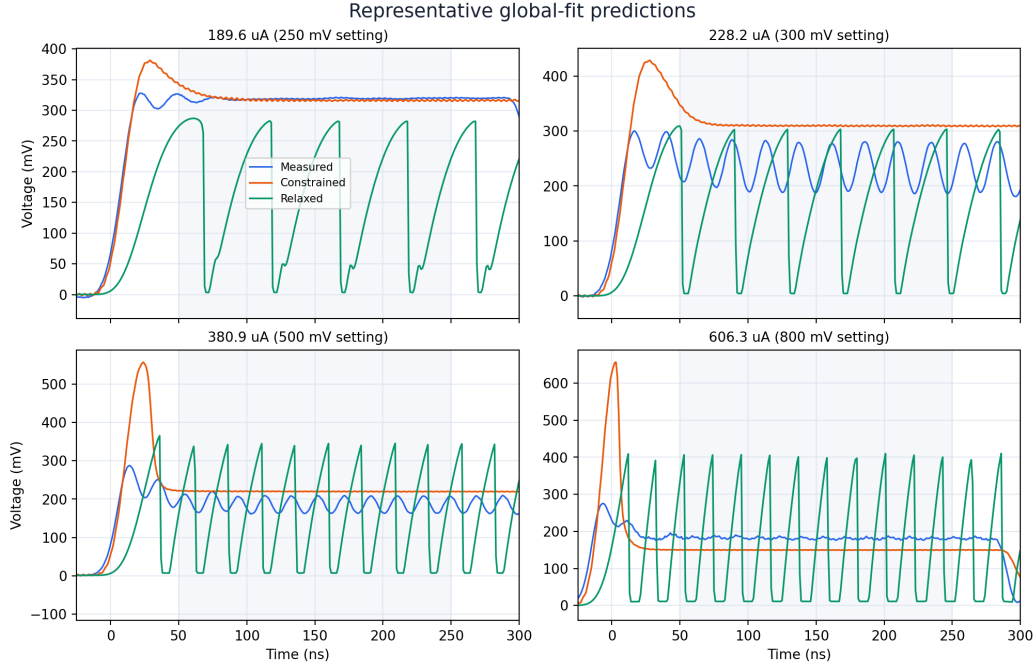


Figure 14: Representative oscillation-priority predictions. Sustained cycles replace the transient-only behavior of the balanced fit, but voltage magnitude and waveform shape remain quantitatively incorrect.

At $\Delta t = 0.025$ ns, the same ten currents oscillate over the identical current interval; the predicted frequency range remains 25.6–50.0 MHz. The classification result is therefore timestep-stable. It is nevertheless a diagnostic, not a new parameter measurement: recovering the window requires C about 35 times its timing upper bound and moves every fitted parameter outside its independent interval. The model contains the mechanism needed for collective switching, but the measured frequency, amplitude, and independently inferred parameter scales cannot yet be satisfied simultaneously. This strengthens the case for including the real TIA/load impedance or a dynamic switching law. The differential-evolution and local-refinement budgets both reached their configured limits, so this is the best reproducible candidate found, not proof of the unique global optimum.

11 Oscillation-amplitude tuning

The classification-first fit recovers the oscillation window, but its voltage cycles are too large. We therefore repeated the same eight-parameter search with two explicit amplitude terms. The absolute term compares peak-to-peak voltages directly, while

$$J_{\text{relamp}} = \left[\ln \left(\frac{V_{\text{pp,pred}} + 3 \text{ mV}}{V_{\text{pp,meas}} + 3 \text{ mV}} \right) \right]^2 \quad (21)$$

penalizes their ratio, so small and large measured cycles matter comparably. The 3 mV floor only keeps the logarithm finite near zero. This term is evaluated only for experimentally oscillatory records. The objective used weights of 2500 for classification and false positives, 10 for relative amplitude, 1 for absolute amplitude, 0.5 for frequency, 0.1 each for waveform, mean voltage, and spectrum, 1 for persistence, and 0.02 for the turn-on edge.

The search used 0.05 ns steps, and the selected vector was then evaluated at 0.025 and 0.0125 ns. Table 13 compares both diagnostic solutions using the same measured traces and the 0.025 ns evaluation.

Quantity	Classification-first	Amplitude-priority
Correct classifications	21 of 22	19 of 22
False positives	0	1
Missed oscillatory records	1	2
Amplitude records compared	10	9
Median predicted/measured V_{pp}	$7.15\times$	$2.59\times$
V_{pp} mean absolute error	280.1 mV	88.5 mV
V_{pp} root-mean-square error	285.7 mV	90.1 mV
Predicted frequency interval	25.6–50.0 MHz	27.8–62.5 MHz

Table 13: Amplitude–classification tradeoff at $\Delta t = 0.025$ ns. Each column’s amplitude statistics use records that oscillate in both experiment and that fit.

The amplitude-priority fit falsely oscillates at 189.6 μA and misses the 532.9 and 606.3 μA records. The same 19/22 classification is obtained at 0.0125 ns, so this tradeoff is not removed by halving the timestep. Its fitted values are $C = 15.672$ pF, $C_{\text{th}} = 0.007770$ pJ/K, $S_e = 0.0009460$ mW/K, $T_0 = 318.761$ K, $T_c = 337.904$ K, $w = 12.123$ K, $\beta = 0.20749$ K^{−1}, and $\gamma = 1.45177$. All except γ lie outside their independent intervals.

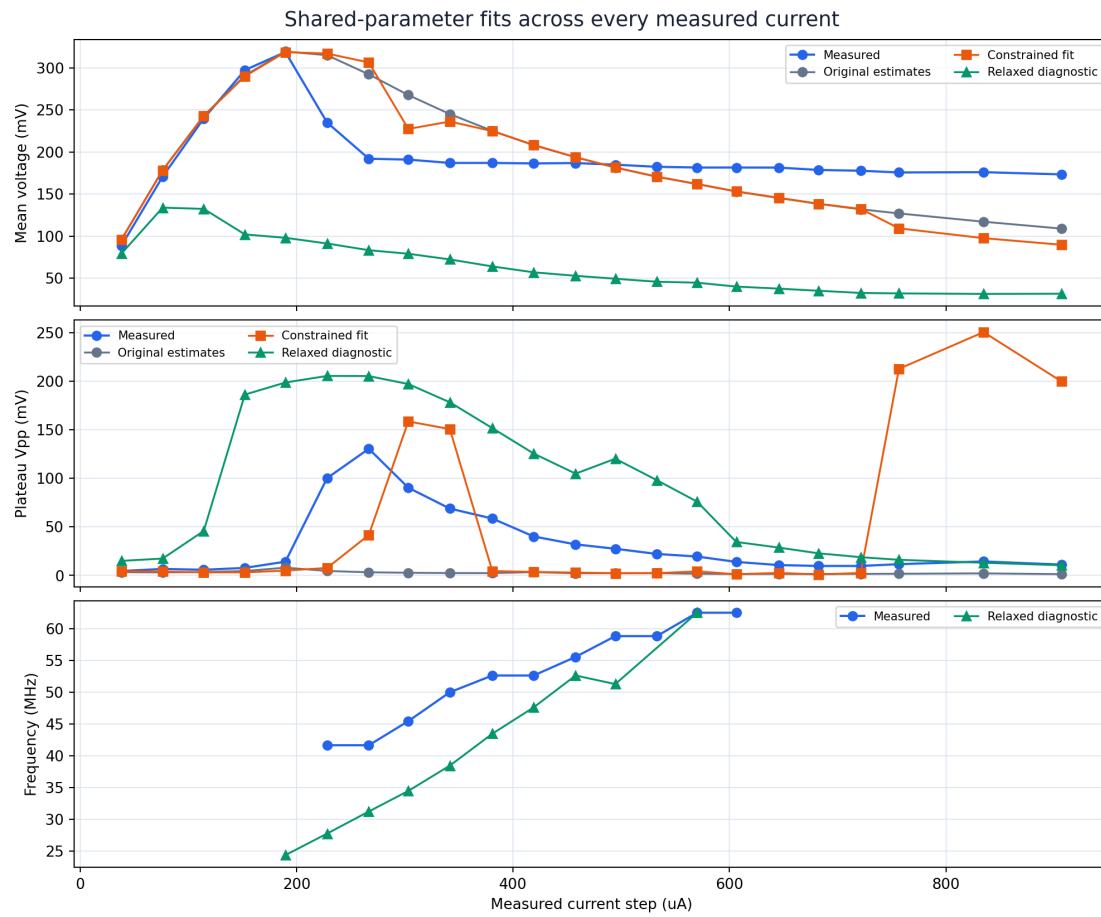


Figure 15: Amplitude-priority operating behavior. Voltage cycles and the upper frequency range move toward experiment, but the oscillation window becomes less accurate.

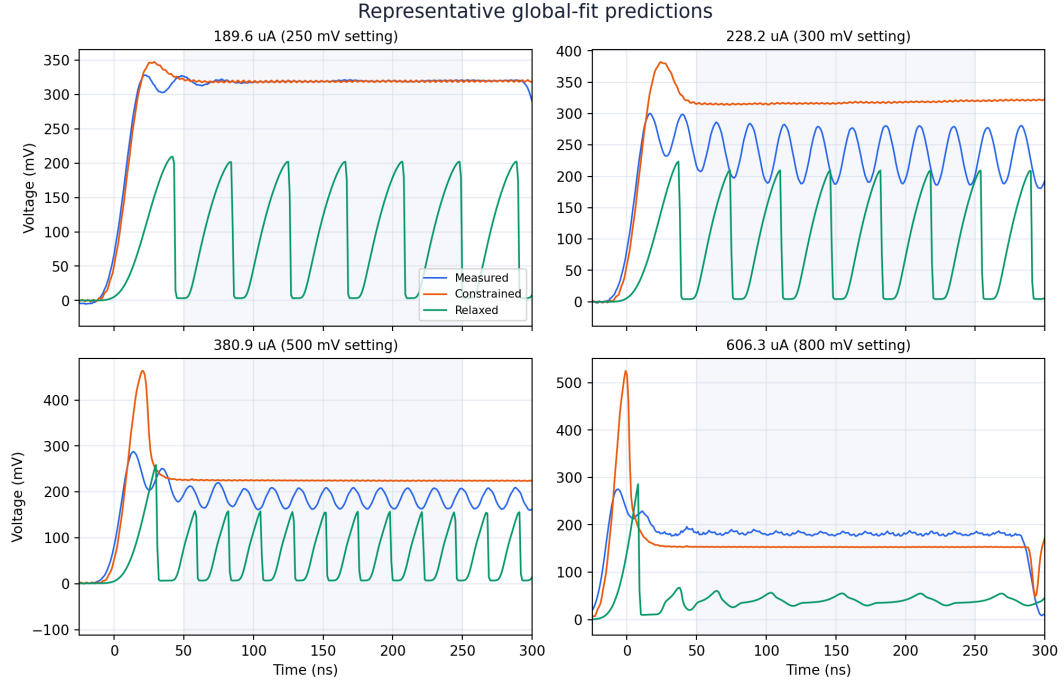


Figure 16: Representative amplitude-priority predictions at 0.025 ns. The cycles remain too large, and the 189.6 μ A false positive makes the tradeoff visible.

This search therefore improves amplitude substantially but does not provide one shared physical parameter vector that also preserves the measured oscillation window. Local moves within the classification basin produced only small amplitude reductions before classifications changed. This sharp observed tradeoff suggests missing amplitude-setting physics rather than a simple optimizer failure. In the ideal-current lumped model, a complete hysteretic transition produces a voltage span set approximately by $I \Delta R$. Partial or spatially nonuniform switching, or the actual TIA/load response, can reduce the measured span without requiring the same changes that move the thermal oscillation boundaries. These are the next model extensions to test. Both optimization stages exhausted their finite budgets, so the result is a reproducible Pareto endpoint rather than proof of a unique global optimum.

A Reproduction

The validation run is archived as

```
20260817_100102_current-step-with-a-nonzero-metallic-voltage-val_6765e0.
```

It can be reproduced from the repository root with

```
neuristor simulate current --config experiments/current/nonzero_voltage_valley.toml
```

The archived bundle contains the resolved configuration, traces, metrics, figure, and repository provenance. A fresh execution produced identical metrics and byte-identical trace data.

The resistance–temperature figure and synchronized GIF are regenerated from the same archived traces with `neuristor runs visualize`.

The sample resistance fit is archived as

```
20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9.
```

It is reproduced with

```
neuristor fit resistance --data data/experimental/100425_chip1_gap3.tsv --method
major-loop --bootstrap-samples 1000
```

The environmental-conductance analysis is archived as

```
20260817_153807_environmental-thermal-conductance-estimate_761640.
```

It is reproduced with

```
neuristor analyze conductance --data data/experimental/tia_current_sweep
--resistance-preset presets/resistance_100425_chip1_gap3.json
--resistance-bootstrap
public_jobs/20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9/parameter_bootstrap.csv
--ambient-K 314.4 --ambient-interval-K 314.25,314.55
```

The thermal-capacitance analysis is archived as

```
20260828_112314_thermal-capacitance-estimate-with-conservative-0_aa2469.
```

It is reproduced with

```
neuristor analyze thermal-capacitance --data data/experimental/tia_current_sweep
--resistance-preset presets/resistance_100425_chip1_gap3.json
--resistance-bootstrap
public_jobs/20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9/parameter_bootstrap.csv
--conductance-mW-per-K 0.003675126546984294
--conductance-bootstrap
public_jobs/20260817_153807_environmental-thermal-conductance-estimate_761640/conductance_bootstrap.csv
--ambient-K 314.4 --electrical-capacitance-pF 0.39
--selected-drives-mV 100,150,200 --fit-window-ns 15,35
```

The blind model comparison and capacitance study are archived as

```
20260829_100718_specimen-model-prediction-versus-measured-current_eefab7.
```

They are reproduced together with

```
neuristor analyze model-validation --config
experiments/current/specimen_model_validation.toml
```

The global shared-parameter inference is archived as

```
20260829_105704_global-specimen-parameter-inference-from-all-current_8f12d6.
```

It is reproduced with

```
neuristor analyze fit-waveforms --config
experiments/current/specimen_waveform_inference.toml
```

The bundle contains the resolved objective and bounds, every fitted parameter, per-trace metrics and predictions, held-out validation, optimization histories, figures, and the three-step convergence check.

The oscillation-priority inference is archived as

```
20260829_160147_oscillation-priority-global-specimen-parameter-inference_ac1c5e.
```

It is reproduced with

```
neuristor analyze fit-waveforms --config
experiments/current/specimen_oscillation_inference.toml
```

Its bundle records the classification-first objective, four-segment persistence metric, parameter search, withheld-current results, per-trace waveforms, and 0.1–0.025 ns convergence audit.

The amplitude-priority inference is archived as

```
20260829_210212_amplitude-tuned-oscillation-priority-specimen-inference_85526c.
```

It is reproduced with

```
neuristor analyze fit-waveforms --config
experiments/current/specimen_oscillation_amplitude_inference.toml
```

Its bundle contains the amplitude-aware objective, all fitted traces and per-trace metrics, parameter table, optimization history, figures, and the 0.1–0.0125 ns convergence audit.

The historical capacitance shortcut is preserved for provenance in

```
20260817_134254_lab-current-trace-parameter-estimates_4223e0.
```

Its `electrical_capacitance.csv` contains the four values reconstructed in Table 6. The bundle is retained as evidence of how 19.8 pF was obtained, but that estimate is superseded by the full current-balance analysis in Section 5.

References

- [1] Y.-H. Zhang, C. Sipling, E. Qiu, I. K. Schuller, and M. Di Ventra, “Collective dynamics and long-range order in thermal neuristor networks,” *Nature Communications*, vol. 15, art. 6986, 2024. doi:10.1038/s41467-024-51254-4.
- [2] A. Gildor, S. Hodisan, S. Kvatinsky, and Y. Kalcheim, “Harnessing the VO₂ Phase Transition for Automatic Gain Control in Transimpedance Amplifiers,” manuscript, arXiv:2604.04594v1, 2026.